

768  
Count of outcome

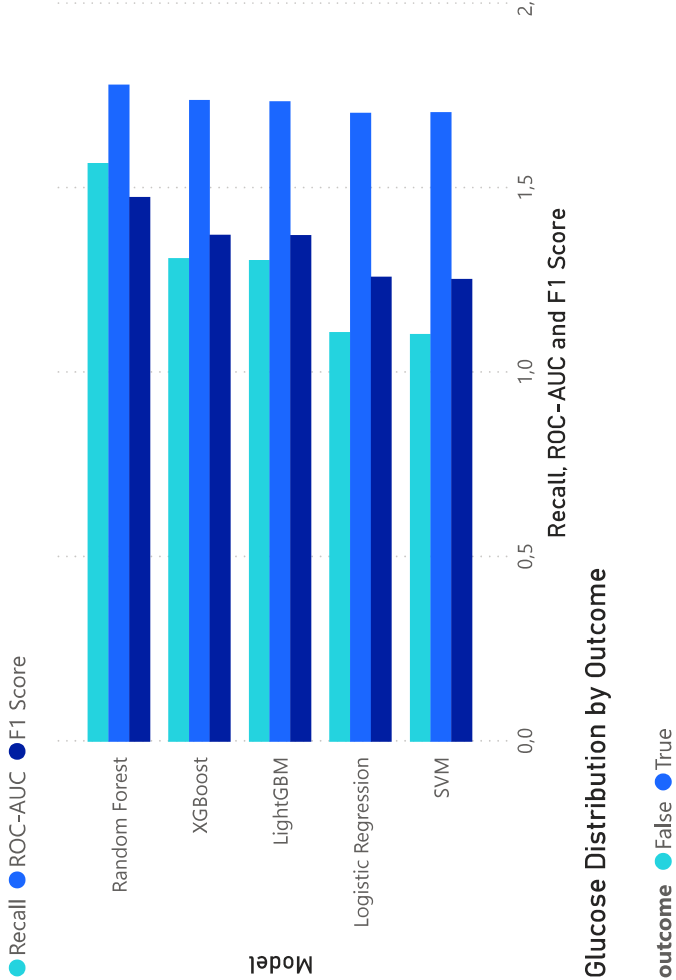
35%  
"% Diabetic"

69%  
Best Recall

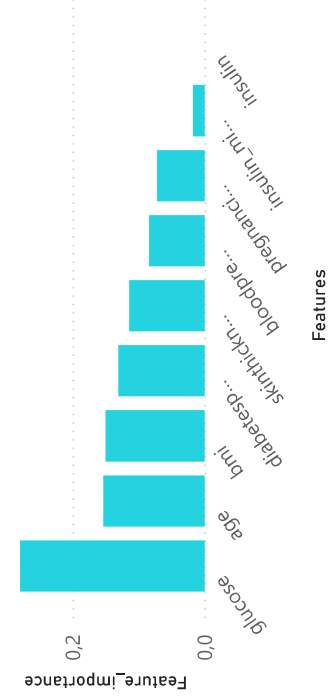
67%  
Best F1 Score

87%  
Best ROC-AUC

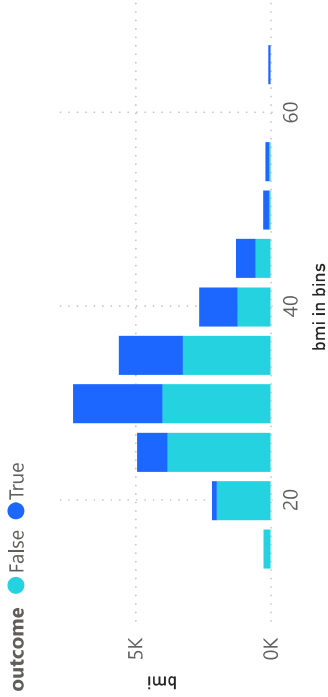
Recall, ROC-AUC and F1 Score by Model



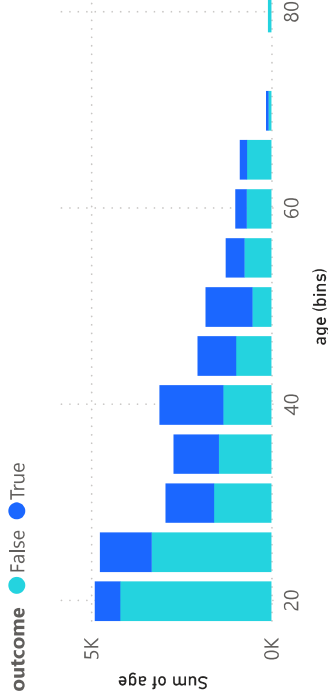
Feature Importance



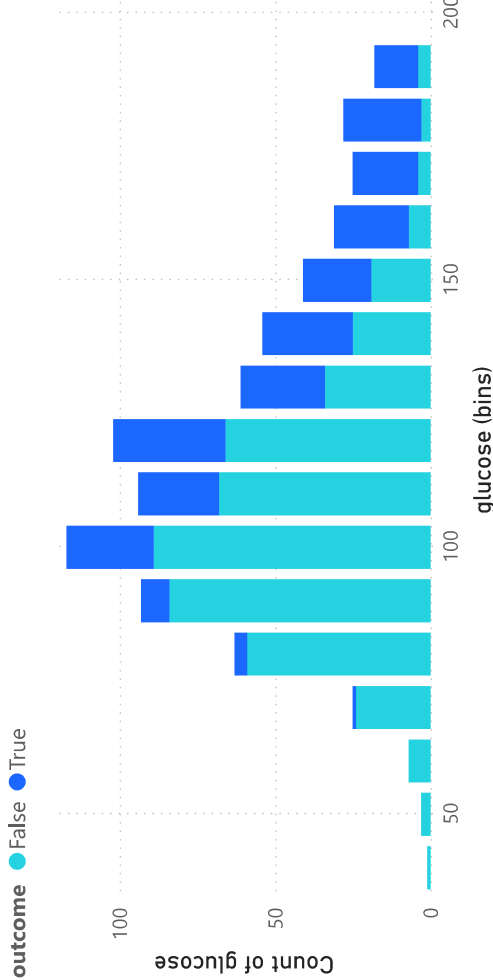
BMI Distribution by Outcome



Age Distribution by Outcome



Glucose Distribution by Outcome



| bmi   | glucose | age | actual_outcome | prob_Random_Forest | prob_LightGBM | pred_Random_Forest | pred_LightGBM |
|-------|---------|-----|----------------|--------------------|---------------|--------------------|---------------|
| 25,60 | 139,00  | 22  | True           | 0,16               | 0,12          | False              | False         |
| 26,40 | 129,00  | 28  | True           | 0,22               | 0,14          | False              | False         |
| 26,50 | 106,00  | 29  | True           | 0,15               | 0,09          | False              | False         |
| 27,10 | 119,00  | 33  | True           | 0,47               | 0,35          | False              | False         |
| 27,60 | 92,00   | 44  | True           | 0,47               | 0,18          | False              | False         |
| 28,40 | 120,00  | 22  | True           | 0,36               | 0,17          | False              | False         |
| 29,00 | 85,00   | 32  | True           | 0,34               | 0,20          | False              | False         |
| 29,70 | 117,00  | 30  | True           | 0,36               | 0,28          | False              | False         |
| 30,50 | 128,00  | 25  | True           | 0,52               | 0,44          | True               | False         |
| 31,60 | 144,00  | 25  | True           | 0,52               | 0,43          | True               | False         |
| 32,00 | 123,00  | 35  | True           | 0,63               | 0,53          | True               | True          |
| 32,00 | 128,00  | 33  | True           | 0,76               | 0,66          | True               | True          |
| 32,30 | 116,00  | 35  | True           | 0,72               | 0,60          | True               | True          |
| 32,70 | 140,00  | 45  | True           | 0,85               | 0,66          | True               | True          |
| 32,80 | 133,00  | 45  | True           | 0,68               | 0,56          | True               | True          |
| 32,90 | 124,00  | 30  | True           | 0,56               | 0,42          | True               | False         |
| 32,90 | 132,00  | 23  | True           | 0,41               | 0,28          | False              | False         |
| 33,30 | 125,00  | 28  | True           | 0,49               | 0,39          | False              | False         |
| 33,60 | 104,00  | 22  | True           | 0,12               | 0,10          | False              | False         |
| 34,40 | 132,00  | 44  | True           | 0,71               | 0,58          | True               | True          |
| 34,50 | 117,00  | 40  | True           | 0,65               | 0,54          | True               | True          |
| 34,60 | 138,00  | 21  | True           | 0,46               | 0,37          | False              | False         |
| 34,80 | 109,00  | 26  | True           | 0,32               | 0,26          | False              | False         |
| 35,60 | 97,00   | 52  | True           | 0,35               | 0,25          | False              | False         |
| 36,00 | 121,00  | 25  | True           | 0,31               | 0,22          | False              | False         |
| 36,50 | 128,00  | 37  | True           | 0,76               | 0,66          | True               | True          |
| 36,60 | 107,00  | 25  | True           | 0,26               | 0,23          | False              | False         |
| 36,60 | 143,00  | 51  | True           | 0,76               | 0,56          | True               | True          |
| 37,40 | 95,00   | 24  | True           | 0,14               | 0,10          | False              | False         |
| 37,90 | 146,00  | 28  | True           | 0,58               | 0,46          | True               | False         |

